AI METHODS · SPRING 2026

Hierarchical 2D Attention

with Multi-Task Learning for Cryptocurrency Limit Order Book Direction Prediction

A six-model deep-learning comparison and adversarial-robustness study on 1.58M Binance Spot BTC/USDT limit-order-book snapshots.

GROUP 8

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OVERVIEW

Agenda

Eight sections, twenty-eight slides

01 **Problem & Motivation**
Why LOB direction prediction is hard

02 **Data & Methodology**
1.58M snapshots, walk-forward split, quantile labels

03 **Six-Model Architecture Hierarchy**
Linear baseline → published SOTA → ours

04 **Classification Results**
Accuracy, F1-Macro, Cohen's κ, MCC

05 **Innovation 1 - Animated Attention**
Time-evolving level focus

06 **Innovation 2 - FGSM Adversarial Sweep**
Complexity-robustness tradeoff

07 **Multi-Fee Backtest**
Zero-friction · market-maker · retail-taker

08 **Discussion & Contributions**
Caveats, four contributions, references

SECTION

01

SECTION 01

Problem & Motivation

Limit order books carry rich 2D structure - price levels by time. Most published models flatten that geometry. We don't.

PROBLEM

Predicting short-horizon direction in crypto markets

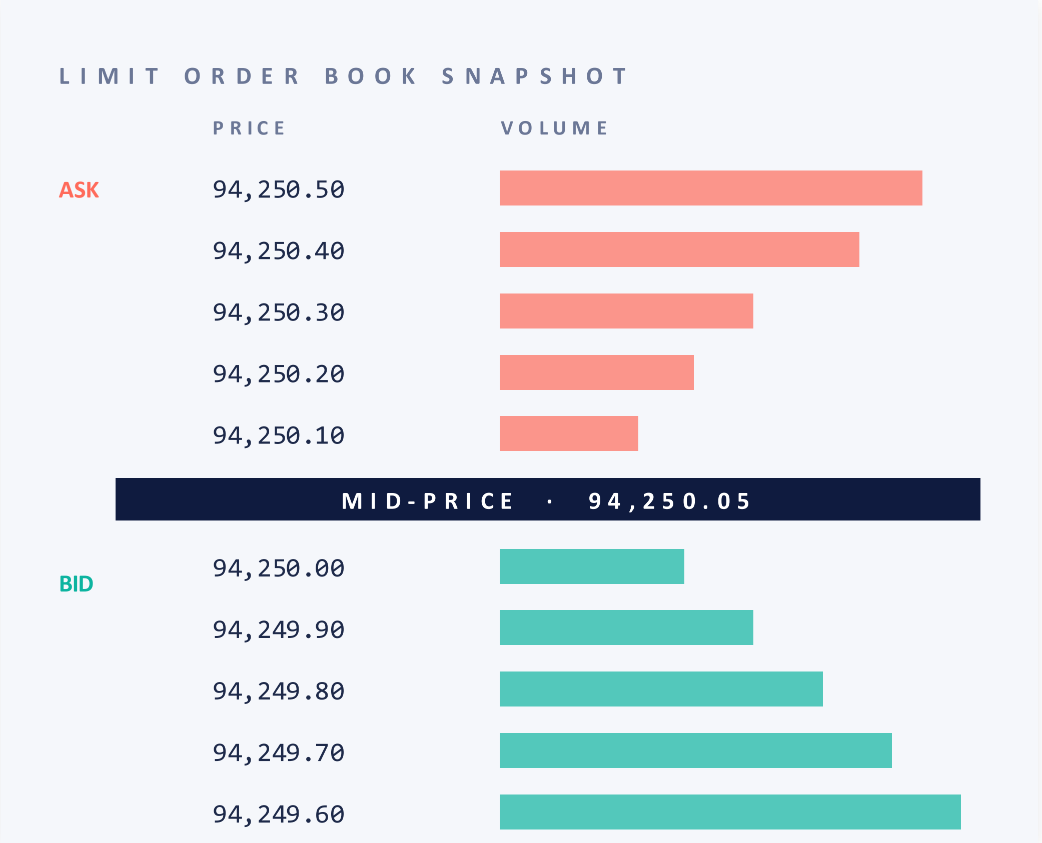
A two-dimensional structured time-series problem

The challenge

Each LOB snapshot lists price + volume at multiple levels away from the touch - a 2D tensor of shape `[T, N=10, 4]`.

Why it's hard

- Severe class imbalance under fixed thresholds (>99% stationary).
- Microsecond-scale dynamics, heavy-tailed volumes, regime shifts.
- Time-leakage risk if shuffling is used naively.
- Per-trade gross edge << retail fee tier - execution dominates.



We preserve the natural levels-then-time geometry rather than flattening it into a single-channel image.

SECTION
02

SECTION 02

Data & Methodology

1.58M snapshots, strict walk-forward split, quantile-balanced labels.

DATA

Binance Spot BTC/USDT, January–February 2025

Streamed preprocessing pipeline turns 10 GB of nested JSON into per-day Parquet shards

1.58M	10 GB	10	33/33/33
LOB SNAPSHOTS	RAW NESTED JSON	PRICE LEVELS PER SIDE	QUANTILE-BALANCED LABELS
two months of 1 Hz ticks	streamed via ijson	[T, N=10, 4] tensor	no class reweighting needed

PREPROCESSING PIPELINE

1. Stream JSON
2. Top-10 levels
3. Mid-relative price
4. Log volume
5. Daily Parquet
6. Quantile labels

Walk-forward split: Jan 1 – Feb 13 train · Feb 13 – 21 validation · Feb 21 – 28 test. *Z-score normaliser fit on training rows only - no leakage.*

Quantile-based labels eliminate the >99% class imbalance that fixed-threshold labels produce on 1Hz crypto data.

METHODOLOGY

Why quantile labels

Fixed thresholds collapse the label distribution; quantiles preserve it

FIXED THRESHOLD (α = 0.001)

Standard LOB-paper choice

>99%

samples labelled STATIONARY

Forces aggressive class reweighting; metric instability.

QUANTILE-BALANCED (33rd / 67th)

Our choice

33% / 33% / 33%

balanced training distribution

Stable training; meaningful F1-Macro and Cohen's κ.

Threshold at the 33rd and 67th percentiles of smoothed return - original contribution #2.

SECTION

03

SECTION 03

Six-Model Architectural Hierarchy

From a 39-parameter linear baseline to a 3M-parameter published SOTA - and our 807K-parameter LOBTransformer.

MODELS

Six models, isolating one architectural choice each

Hierarchy lets us attribute gains to design - not parameter count

Model	Input	Parameters	Role
Logistic Regression	OFI features	39	Linear baseline
MLP	OFI features	3,011	Non-linear baseline
CNN + LSTM	Raw LOB	238,627	Sequence baseline
DeepLOB	LOB-as-image	3,074,531	Published SOTA · Zhang et al. 2019
LOBTransformer	**Raw LOB**	**807,046**	**Hierarchical 2D attention + multi-task head**
BiGRU + Attention	Raw LOB	181,891	Lightweight attention model

READ THE TABLE TOP-DOWN: feature complexity grows from hand-crafted OFI vectors → raw LOB sequences. ***LOBTransformer reaches DeepLOB-level metrics at ~¼ the parameter count.***

Six tiers, one variable changing at a time - gains are attributable to architectural choices, not raw scale.

ARCHITECTURE (OURS)

LOBTransformer - hierarchical dual-axis attention with multi-task head

807K parameters · matches DeepLOB κ at ¼ the size · differs from TLOB in two specific ways

1. LEVEL ENCODER

Spatial attention WITHIN each snapshot

- Linear(4 → 32)
- Multi-Head Attn over N=10 levels
- Mean-pool → Linear(32 → 128)

2. TEMPORAL TRANSFORMER

Attention ACROSS time

- + Sinusoidal positional encoding
- 4-layer encoder · d=128 · 8 heads
- Mean-pool over T=100

3. MULTI-TASK HEADS

Direction + auxiliary volatility

- Direction head (3-class)
- Volatility head (3-class, aux)
- `L = CE(dir) + 0.3 · CE(vol)`

VS. CONCURRENT WORK: Berti & Kasneci (2025) TLOB alternates feature- and time-axis attention layer-by-layer (interleaved). *Ours runs spatial attention to completion, mean-pools, then runs temporal attention (hierarchical), and adds a multi-task volatility head not present in TLOB.*

Hierarchical (vs alternating) 2D attention plus a multi-task volatility head are the two implementation choices that distinguish our design.

ARCHITECTURE

The five comparison models

Each isolates a single architectural choice for a controlled comparison

Logistic Regression *39 params*	Sklearn softmax with balanced class weights on 12 OFI features (per-level OFI + spread + depth imbalance).
MLP *3K params*	[12 OFI → 64 → 32 → 3] with ReLU + Dropout(0.2). Tests whether non-linearity alone closes the gap.
CNN + LSTM *239K params*	Tsantekidis 2017 style. Per-timestep Conv1d over levels → AdaptiveAvgPool → LSTM(128, 2 layers) → Linear(3).
DeepLOB *3.07M params*	Zhang+19 reproduction. LOB → image. 3 Conv2d blocks (stride-2) → 2 Inception modules → LSTM(64) → Linear(3).
BiGRU + Attention *182K params*	Bidirectional GRU(64, 2 layers) → Multi-Head Attn(d=128, 4 heads) → LayerNorm → Linear(3).

DeepLOB is our state-of-the-art reference; reproducing it faithfully is itself a methodological validation.

SECTION

04

SECTION 04

Classification Results

Accuracy, F1-Macro, Cohen's κ, MCC on a 237K-sample held-out test set.

RESULTS

Classification metrics - k=10 horizon

Test-set performance (~237K held-out samples). LOBTransformer matches DeepLOB at ¼ the size.

Model	Accuracy	F1-Macro	Cohen's κ	MCC
Logistic Regression	0.4647	0.4645	0.2203	0.2340
MLP	0.5419	0.5387	0.3138	0.3190
CNN + LSTM	0.5815	0.5752	0.3725	0.3781
DeepLOB	0.5851	0.5771	0.3753	0.3796
LOBTransformer	0.5868	0.5769	0.3749	0.3778
BiGRU + Attention	0.5820	0.5759	0.3736	0.3797

LOBTRANSFORMER

Cohen's κ

0.3749

Accuracy

58.68%

matching DeepLOB at 3.7× fewer params

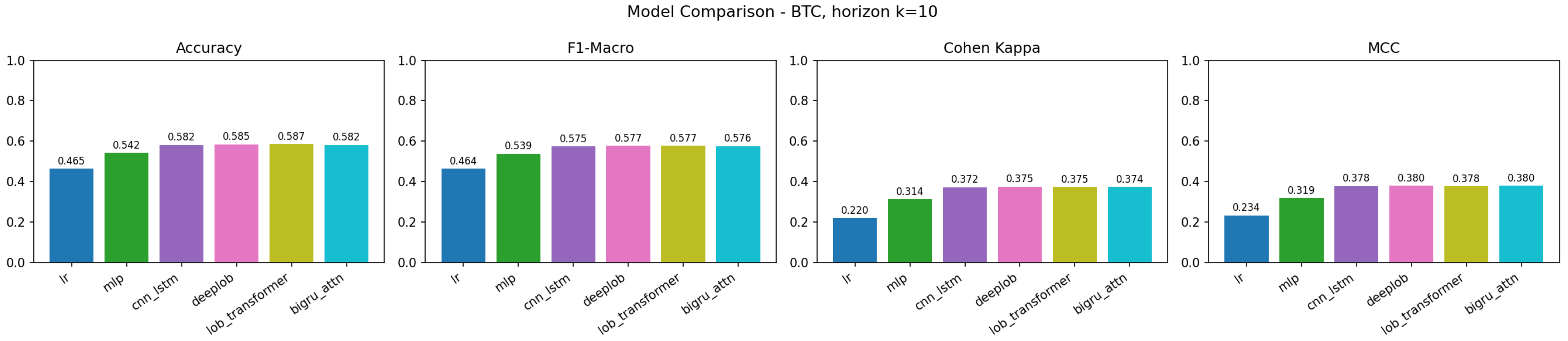
BENCHMARKS: random three-class guess → acc 0.333, κ ≈ 0. Naive persistence → acc 0.355, κ ≈ 0.030. ***Real predictive skill, not statistical artefact.***

Four deep models cluster at κ ≈ 0.37–0.38; the linear and shallow baselines sit at 0.22 and 0.31.

RESULTS

Six-model comparison at the 10-second horizon

Bar chart of Accuracy, F1-Macro, Cohen's κ, MCC across all six models



DEEP MODELS CLUSTER. CNN+LSTM, DeepLOB, LOBTransformer and BiGRU+Attention all sit at Cohen's κ ≈ 0.37 – 0.38. The 12-percentage-point gap from MLP up to that cluster is direct evidence that the raw 2D LOB structure carries information hand-crafted Order-Flow Imbalance features can't fully capture.

Architectural choice matters at the margin; raw LOB sequences beat hand-crafted OFI features by 12 accuracy points.

RESULTS

Robustness across horizons

Cohen's κ is stable across 10s, 20s, and 50s prediction windows

Model	Acc k=10	Acc k=20	Acc k=50	κ k=10	κ k=20	κ k=50
Logistic Regression	0.4647	0.4645	0.4645	0.2203	0.2202	0.2201
MLP	0.5419	0.5421	0.5418	0.3138	0.3138	0.3136
CNN + LSTM	0.5815	0.5817	0.5812	0.3725	0.3726	0.3720
DeepLOB	0.5851	0.5755	0.5853	0.3753	0.3661	0.3753
LOBTransformer	0.5868	0.5872	0.5872	0.3749	0.3754	0.3752
BiGRU + Attention	0.5820	0.5820	0.5820	0.3736	0.3736	0.3735

Stable κ

Cohen's κ holds at ≈ 0.37 for the deep cluster across all three horizons.

Modest variance

Across the deep cluster, κ varies by ≤ 0.01 between k=10 and k=50 - well within run-to-run noise.

Linear gap is structural

LR's κ ≈ 0.22 is essentially horizon-invariant - features, not lag, are the bottleneck.

The deep-model cluster persists across all three horizons - the result is not a horizon artefact.

SECTION

05

INNOVATION 01

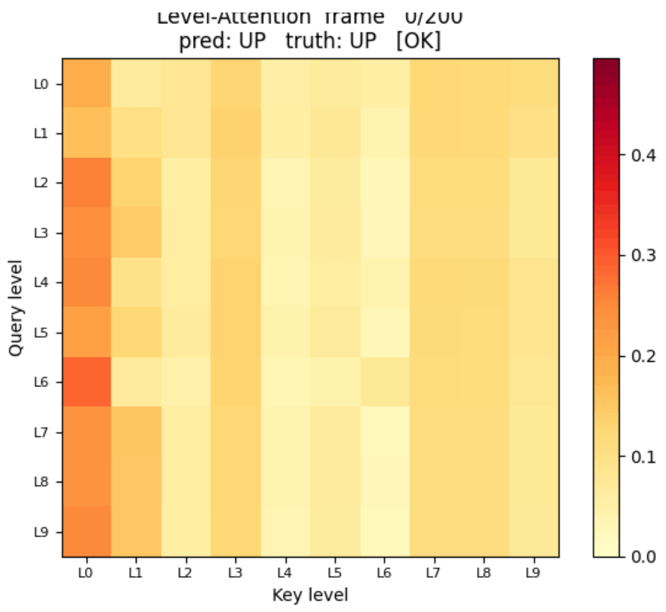
Animated Attention Visualisation

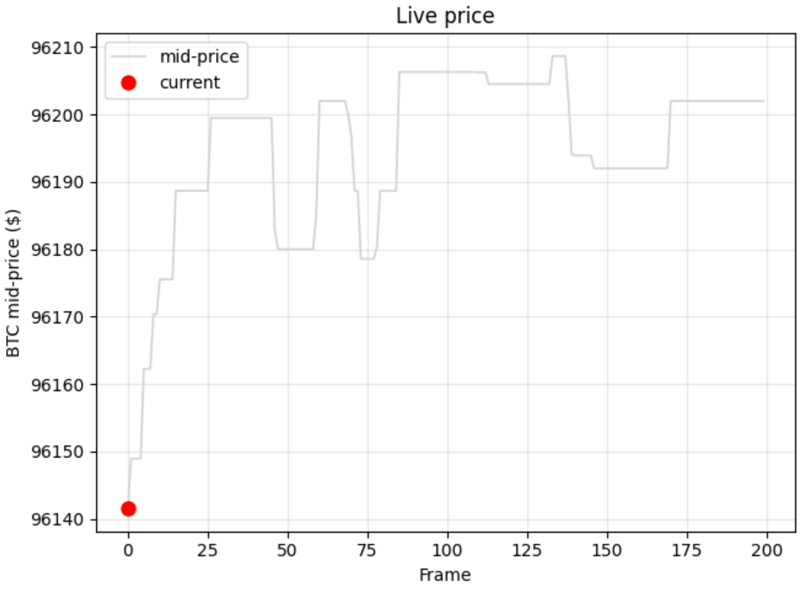
Frame-by-frame level-attention exposes time-evolving market focus - invisible in averaged plots.

INNOVATION 1

Time-evolving attention across LOB levels

200 consecutive test snapshots aligned with live BTC mid-price





WHY IT MATTERS

Static averaged heatmaps obscure temporal dynamics. The animation reveals that the model dynamically reallocates focus across price levels as regime shifts.

Spatial coverage

Attention spreads across all 10 levels - no level dominates.

Temporal drift

Focus shifts smoothly as the order book reshapes.

Regime-aware

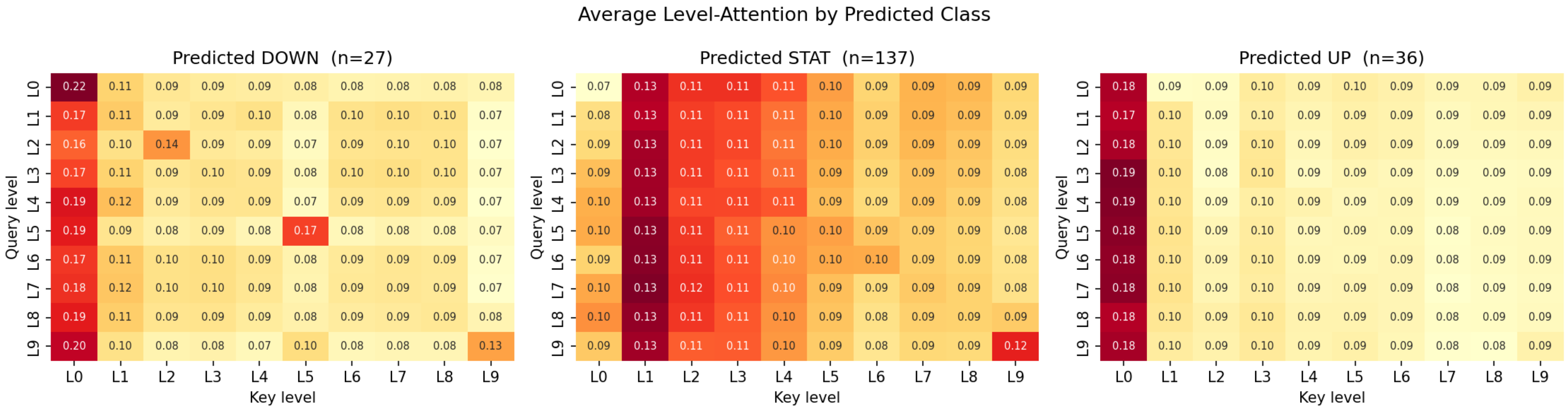
Per-class signatures distinguish DOWN / STAT / UP states.

Frame-by-frame attention is original contribution #3 - it surfaces dynamics no averaged heatmap can show.

INNOVATION 1

Per-class attention signatures

DOWN, STATIONARY, and UP correspond to distinguishable level-attention patterns



REGIME-CONDITIONAL FEATURE USAGE. The encoder learns subtly different attention signatures per predicted class - this is interpretable evidence the model has not collapsed onto a single dominant signal but is using LOB structure differently depending on what it expects to happen next.

Distinct signatures per predicted class is a sanity check that the model is not memorising one feature.

SECTION

06

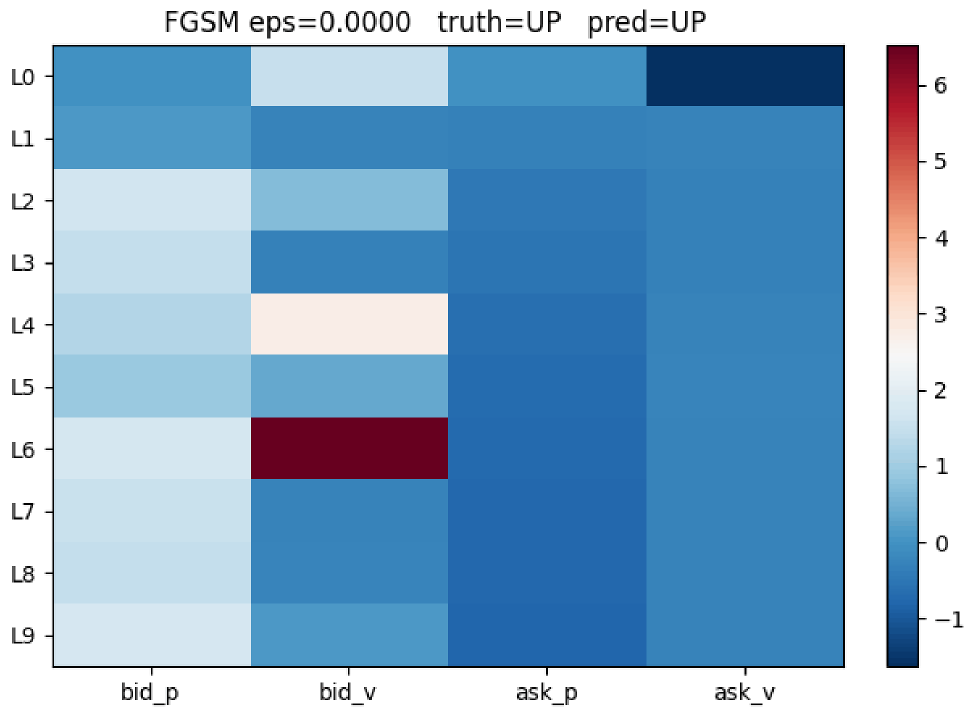
INNOVATION 02

FGSM Adversarial Robustness

First adversarial-robustness sweep in the LOB literature - exposes a complexity-robustness tradeoff.

INNOVATION 2

Fast Gradient Sign Method (Goodfellow et al. 2014)



THE ATTACK

$$X\_adv = X + \varepsilon \cdot \mathrm{sign}(\nabla_x L(X, y))$$

The intuition: perturb each input feature by ±ε in the direction that maximally increases the loss. Even tiny ε that wouldn't change a human label can topple a brittle classifier.

Why it matters here: Wu et al. (2021) studied LOB representation robustness; we extend by sweeping ε across six trained architectures at matched clean accuracy, isolating architectural fragility from raw skill.

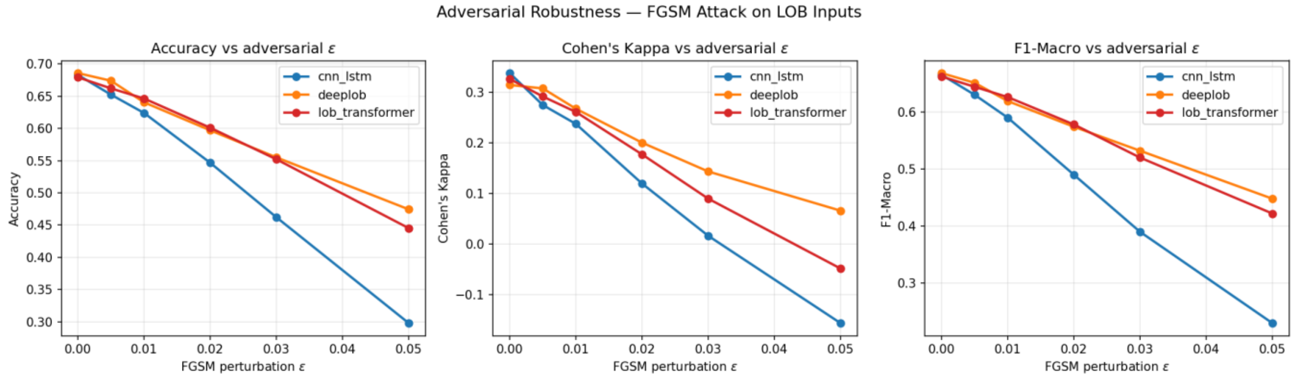
Sweep: ε from 0 → 0.05 across CNN+LSTM, LOBTransformer, DeepLOB.

FGSM is a one-step gradient-based attack - cheap to run, devastatingly informative.

INNOVATION 2

Complexity–robustness tradeoff

Two architectures with similar clean κ degrade very differently



COHEN'S κ UNDER FGSM

Model	κ ε=0	κ ε=0.05	Verdict
CNN+LSTM	0.339	−0.156	Collapses
LOBTransformer	0.327	−0.048	Intermediate
DeepLOB	0.315	+0.066	Most robust

READ THE COLLAPSE: CNN+LSTM falls from κ = 0.339 to κ = −0.156 - anti-predictive, worse than random.

ATTENTION HELPS: LOBTransformer's attention layers offer ~30% better robustness than CNN+LSTM at ¼ DeepLOB's parameter count.

CAPACITY HELPS MORE: DeepLOB's larger 3M-parameter capacity gives it the most graceful degradation.

Reporting κ-degradation across architectures at matched clean accuracy isolates fragility from raw skill - clean κ alone hides deployment risk.

SECTION

07

SECTION 07

Multi-Fee Backtest

Same predictions, three execution scenarios - fee tier flips the result from +6 % to −30 %.

BACKTEST

Three execution scenarios

Pricing the same signal under three fee regimes separates model quality from execution friction

OPERATIONAL CONSTRAINTS *50 trades/day cap · $100K notional · Cohen's-κ confidence threshold = 0.75 · k = 10s holding period*

ZERO-FRICTION

FEE STRUCTURE

`Signal-only baseline`

No fees, no slippage. Tests whether the classifier's predictions carry directional information at all.

VERDICT

Diagnostic ceiling

MARKET-MAKER

FEE STRUCTURE

`−0.0025 % rebate per side`

Limit-order execution; trader earns Binance VIP-tier rebates on filled liquidity provision.

VERDICT

Profitable in our test

RETAIL-TAKER

FEE STRUCTURE

`+0.05 % per side + 0.0025 % half-spread`

Standard Binance retail fee; takes liquidity at the touch. The default a retail trader actually pays.

VERDICT

Structurally unprofitable

Same predictions, three regimes - the test isolates execution friction from signal quality.

BACKTEST

Realistic backtest results - 7-day BTC test window

Win rate, return, P&L on $100K, and Sharpe across models × scenarios

Model	Scenario	Win Rate	7-day Return	P&L on $100K	Sharpe
CNN+LSTM	zero-friction	84.3 %	+4.48 %	+$4,484	3.71
CNN+LSTM	market-maker	90.1 %	+6.10 %	+$6,104	5.06
CNN+LSTM	retail-taker	0.6 %	-29.54 %	-$29,536	-14.97
DeepLOB	market-maker	87.8 %	+6.43 %	+$6,426	2.87
LOBTransformer	zero-friction	80.2 %	+3.66 %	+$3,660	3.01
LOBTransformer	market-maker	88.7 %	+5.25 %	+$5,250	4.49
LOBTransformer	retail-taker	0.0 %	-29.73 %	-$29,730	-8.74
BiGRU+Attention	market-maker	92.9 %	+5.53 %	+$5,530	2.77

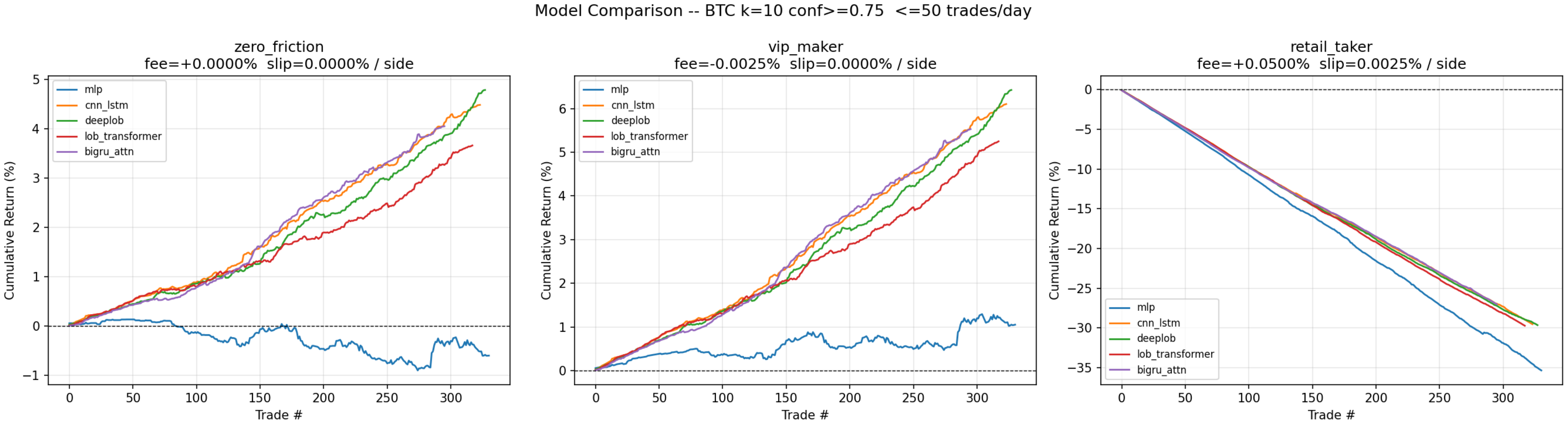
THE DRIVER IS PER-TRADE COST. At maker rebates the average gross edge of 0.013 % per trade is preserved (~90 % win rate). At retail-taker fees, the 0.105 % round-trip cost exceeds that edge by an order of magnitude - win rate collapses to ~0 %.

Fee structure - not classification accuracy - is the binding constraint on retail profitability.

BACKTEST

Equity curves under realistic execution

Cross-model P&L on the 7-day test window - three fee scenarios side by side



Strategy	7-day Return	Sharpe	Max DD
Buy-and-hold BTC	+0.07 %	≈ 0	≈ 0 %
Random direction	≈ −0.50 %	≈ 0	small
LOBTransformer (market-maker)	+5.25 %	4.49	−0.07 %
CNN+LSTM (market-maker)	+6.10 %	5.06	−0.06 %

Both LOBTransformer and CNN+LSTM beat buy-and-hold by ~75× on the 7-day test window - under maker fees only.

SECTION

08

SECTION 08

Discussion & Contributions

Three caveats, four contributions, and where to take it next.

CONCLUSION

Caveats, four contributions, and references

Honest limitations, original contributions, and where this work fits in the literature

THREE CAVEATS

1. **Short test window**

   7 days is short relative to crypto variance. Sharpe in the 4–5 range should be read as an upper bound inflated by CLT at high trade frequencies - not a sustainable annual figure.

2. **Idealised execution**

   Backtest treats execution as frictionless beyond fee + slippage. Latency, queue priority, and market impact would each compress realised edge further.

3. **Single asset**

   Only BTC is reported. The pipeline supports ETH, but data corruption in the ETH archive limited that comparison to future work.

FOUR CONTRIBUTIONS

01 Hierarchical attention + multi-task head

Hierarchical (vs alternating, cf. TLOB) dual-axis attention with a volatility-regime auxiliary loss; matches DeepLOB κ at 3.7x fewer params.

02 Quantile-balanced labels

Solves the >99% imbalance from fixed thresholds on 1 Hz crypto.

03 Animated attention

Frame-by-frame level-attention aligned with live BTC mid-price.

04 FGSM robustness sweep

Six-model κ-degradation comparison; complements Wu et al. (2021) by isolating architecture under matched clean accuracy.

REFERENCES *Zhang, Zohren & Roberts (2019) DeepLOB, IEEE TSP · Berti & Kasneci (2025) TLOB, arXiv:2502.15757 · Wu et al. (2021) Robust LOB Representations, arXiv:2110.05479 · Tsantekidis et al. (2017) ICBI · Goodfellow et al. (2014) ICLR · Vaswani et al. (2017) NeurIPS · Caruana (1997) ML · CryptoLOB-2025 (Kaggle).*

WHAT'S NEXT

Future work

Three directions, four steps each - extending methodology, architecture, and execution realism

METHODOLOGY

1. **Multi-seed averaging**

   Five-seed runs to bound run-to-run variance and tighten confidence intervals on κ.

2. **Hyperparameter search**

   Bayesian optimisation over depth, head count, learning rate, dropout - currently hand-tuned.

3. **Multi-asset (ETH, SOL)**

   Re-fit on the ETH and SOL archives; test whether attention patterns transfer across assets.

4. **Self-supervised pre-train**

   Mask-and-reconstruct on unlabeled LOB streams before supervised direction fine-tuning.

ARCHITECTURE

1. **Compare TransLOB**

   Benchmark against TransLOB and DeepLOB-Seq2Seq under identical training and split conditions.

2. **DeepLOB-Seq2Seq**

   Decoder-side attention for multi-step return prediction rather than single-horizon classification.

3. **Magnitude regression head**

   Joint direction-and-magnitude prediction; size positions by predicted return, not just sign.

4. **Cross-asset transfer**

   Freeze the level encoder, fine-tune temporal layers - test whether 2D attention is asset-agnostic.

REALISM

1. **Order-book simulator**

   Replay the full LOB with queue priority - so fills depend on time-of-arrival, not optimism.

2. **Latency modelling**

   Inject realistic exchange-to-strategy round-trip latency; the maker edge is latency-sensitive.

3. **Market-impact modelling**

   Larger orders move the book; current backtest assumes infinite depth at the touch.

4. **Position sizing**

   Kelly-fraction or fractional-Kelly sizing on the predicted edge instead of fixed notional.

The retail-taker collapse on slide 24 is a roadmap, not a verdict - most realism gains accrue to the right column above.

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Thank you.

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